

| **ARIA-NBV: Target- Conditioned RRI + Q_H**

**Rollout data, finite-candidate
planning, advisor alignment**

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Central Research Question

Main Thesis Question

Can ARIA-NBV perform **target-conditioned, RRI-based** multi-step NBV by training a **discrete finite-candidate** value model $Q_{H,\theta}$ that predicts **fixed-horizon cumulative target-specific RRI** and improves **endpoint target reconstruction quality** over a **myopic scorer**?

$$\Delta_0^e = d(C_e(\mathcal{P}_0), \mathcal{M}_e^{\text{GT}}) \quad \Delta_H^e = d(C_e(\mathcal{P}_0 \cup \mathcal{P}_1 \cup \dots \cup \mathcal{P}_H), \mathcal{M}_e^{\text{GT}})$$

$$J_e^{(H)}(\tau) = \frac{\Delta_0^e - \Delta_H^e}{\Delta_0^e + \varepsilon}$$

Gated Research Questions

- RQ1 Method** Which target-RRI objective, RL methodology and offline finite-candidate Q_H formulation are most idiomatic? Fixed vs variable horizon?
- RQ2 Offline** Can offline finite-candidate Q_H recover headroom over a learned myopic scorer, myopically sampled oracle rollouts or even Gumbel-Top- k oracle rollouts?
- RQ3 Repr.** Which actor-visible scene, target, history, visibility, and candidate-view representations improve Q_H under RQ2: semidense geometry, DINO-on-point, EVL/OBB evidence, crop descriptors, S^2 view memory, and candidate frustum/support?
- RQ4 Support** Do realistic mixed candidates and replayable rollout traces provide hard-valid, diverse support, and does scaling this support improve held-out Q_H generalization? How can we scale beyond the GT mesh-labeled dataset?
- RQ5 Online** After offline Q_H , does online training over the same discrete finite-candidate contract improve endpoint gain? Can the multi-step scorer predict invalid actions?
- RQ6 Cont.** Do continuous/hierarchical target-then-pose policies, e.g. actor-critic, yield headroom over the best finite-candidate policy under the same target-RRI objective?

RQ3: Scene / Target / Candidate Tokens

Scene memory stance

- EVL: actor-visible target support and local evidence.
- Semidense/fused points: broader scene memory.
- DINO-on-point features: planned representation ablation, not current cache schema.
- Target-specific RRI and Q_H remain the utility and oracle-evaluation signal.

$$\Phi_t^{\text{scene}} = (X_t^{\text{pt}}, F_t^{\text{DINO@pt}}, E_0^{\text{EVL-local}}, O_t^{\text{pred}}, M_{\text{dir}})$$

$$x_j^{\text{pt}} = \text{concat}(p_j, f_j^{\text{DINO-comp}}, \sigma_j^{-1}, n_j, a_j^{\text{hist}})$$

Queryable target and candidate evidence

Candidate rows query the same actor-visible feature bank through target, frustum, and target-frustum intersection reads:

$$z_e = \text{Pool}_{p_j \in \hat{B}_e} x_j^{\text{pt}}$$

$$z_i^{\text{fr}} = \text{Pool}_{p_j \in \text{Frustum}(q_{t,i})} x_j^{\text{pt}}$$

$$z_{e,i}^{\text{cap}} = \text{Pool}_{p_j \in \hat{B}_e \cap \text{Frustum}(q_{t,i})} x_j^{\text{pt}}$$

scene bank

target token T_e

candidate
rows

Target crops use observed/predicted OBBs. GT crops and meshes remain labels/evaluation only.

RQ3 Detail: Target / View / History Budget

Block	Baseline	Extensions
Target encoding	observed/predicted OBB geometry; class/confidence; projected area; semidense/EVL features; relative target pose	actor-visible crop descriptor ablation; final match/support thresholds and ambiguity policy
View / candidate encoding	pose; target-relative pose; R6D rotation; target bearing/incidence; path cost; strategy provenance; frustum and target-frustum support	query-centric relative bias; Fisher/SCONE overlap channels; held-out candidate-generator stress
History / direction	selected-view history; remaining budget; accumulated geometry; direction moment/novelty around target-local cells	richer counterfactual RGB/SLAM/semantic history unless synthesized later
Validity / provenance	hard mask; invalid reason; valid count; strategy/mixture ids; per-family support and gain diagnostics	row shuffle; duplicate-row; mask isolation; valid-count sensitivity; with/without-strategy tests

Table 1: Actor-visible feature budget and missing ablations before interpreting set-interaction gains.

Compact notation

$$z_e = \varphi(\hat{B}_e, \hat{y}_e, \hat{p}_e, A_e^{\text{proj}}, n_e^{\text{semi}}, n_e^{\text{EVL}}, T_e^{\text{rel}})$$

$$x_{\text{pose}}(q_{t,i}) = \text{concat}(t_{t,i}, R_{t,i}^{6D}, \|t_{t,i} - t_e\|_2^2, \alpha_{t,i}^e, l_{t,i}, c_{t,i}^{\text{strategy}})$$

Boundary: GT OBBs, GT target crops, GT meshes, oracle RRI, and all-candidate GT renders are labels/evaluation only, not V1 actor-visible encodings.

Conservative rule: feature channels and set interaction stay ablations until myopic calibration, masks, replay support, and oracle-rescored selected actions are trusted.

RQ4: Support + Scale Controls

Support contract

Candidate and rollout data are useful only if they cover valid target-relevant decisions, not just many rows.

- Mixed target-centric and exploration families.
- Regenerated successor candidate tables.
- Hard-valid masks plus reason codes for invalid rows.
- Held-out Q_H generalization under fixed target-RRI supervision.
- External data only if comparable mesh/oracle target labels exist.

Invalid candidates are support failures or constraints, never low-RRI examples.

Gate	Check	Purpose
S0	H=1 label profile	target gain is non-flat and valid rows exist
S1	mixed candidate preflight	target-aware support plus exploration
S2	H>1 replay traces	successor tables and selected-depth lineage
S3	stochastic support	avoid fitting only greedy chains
S4	scene-level scale	held-out generalization without split leakage
M0	MLP / DeepSets / Set Transformer	model-control ladder for attribution

Table 2: Support and model controls before interpreting value-model gains.

Do not compare architecture, rollout, or Q_H conclusions across runs that silently change scene, target, candidate, or validity coverage.

RQ2: Offline Finite-Candidate Q_H

Training path

- 1 Train target-conditioned one-step scorer $\hat{r}_\psi^e$ on valid all-candidate labels.
- 2 Use held-out rank, top- k , calibration, and oracle-selected rollouts as evidence gate.
- 3 Fit residual dueling Q_H on selected-action transitions.
- 4 Decode only over hard-valid candidates.

Why this RQ exists: test the actual planning claim after the myopic control and replay support are trusted.

$$Q_{H,\theta} = \hat{r}_\psi^e + V_\theta + A_{\theta,i}^H - \bar{A}_\theta^H, \quad \bar{A}_\theta^H = \frac{1}{|\mathcal{A}_t|} \sum_{j \in \mathcal{A}_t} A_{\theta,j}^H$$

Readable replay contract

all valid candidate labels	myopic scorer $\hat{r}_\psi^e$	calibrated control
selected transitions	successor Q_{t+1}	masked replay rows
residual Q_H	oracle-rescore policy	report η_Q

Masked Double-Q target

$$i^* = \arg \max_{i \in \mathcal{A}_{t+1}} Q_\theta(s_{t+1}^{\text{cf0}}, z_e, q_{t+1,i})$$

$$y_t = r_{t,\text{root}}^e + \gamma(1 - d_t)Q_{\theta-}(s_{t+1}^{\text{cf0}}, z_e, q_{t+1,i^*})$$

Non-expanded candidates train myopic rows only; Q_H backups require selected transitions and successor candidate tables.

RQ2: Offline Q_H Evidence Gates

Matched policy ladder

- π_{rand} : random valid lower reference.
- $\pi_{\text{oracle-1}}$: one-step oracle greedy upper reference.
- $\pi_{\text{oracle-H}}$: bounded oracle lookahead / headroom reference.
- $\pi_{\text{learned-1}}$: learned actor-visible myopic scorer.
- π_Q : offline learned Q_H policy.

Gumbel-Top- k belongs to rollout-support diversity, not the main policy ladder.

Headroom and recovery

$$\Delta_{\text{look}} = J_e^{(H)}(\pi_{\text{oracle-look}}) - J_e^{(H)}(\pi_{\text{oracle-1}})$$

$$\eta_Q = \frac{J_e^{(H)}(\pi_Q) - J_e^{(H)}(\pi_{\text{learned-1}})}{J_e^{(H)}(\pi_{\text{oracle-look}}) - J_e^{(H)}(\pi_{\text{learned-1}}) + \varepsilon}$$

Reported Quantities

- endpoint target gain;
- cumulative target-root gain;
- diagnostic target and scene RRI;
- invalid-action rate and reason distribution;
- path length, runtime, oracle eval count;
- coverage gaps and scene-level splits.

RQ5/RQ6: Online and Continuous Headroom

RQ5: discrete online training

After offline Q_H is stable, test online interaction over the same finite candidate action contract.

- Same target-RRI reward and oracle re-evaluation.
- Same actor-visible state and masks/reasons.
- Compare offline fitted Q_H vs online-updated finite-candidate policy.
- Pass only if endpoint gain or recovered headroom improves under matched budget.

RQ6: continuous hierarchical actions

If finite-candidate evidence is stable, test whether continuous target-then-pose or hierarchical actions have extra headroom.

- Factor target/look-at choice from feasible pose realization.
- Compare against the best finite-candidate policy, not random.
- Keep target-RRI objective and leakage rules unchanged.
- Actor-critic may reuse learned Q_H as initialization or critic signal.

RQ5/RQ6 are gated extension RQs: attempt them only when finite-candidate evidence and compute make the comparison interpretable.

RQ-Gated Roadmap to Next Milestones

M	Gate	Exit
M1	data, cache, oracle	frame/CW90/RRI contract and throughput
M2	one-step baseline	calibration/ranking and scale plan
M3	target RRI	V0/V1 target contract and sharding gate
M4	target scorer	observed/predicted target-conditioned RRI
M5	rollouts and Q_H	headroom, η_Q , oracle-evaluated actions
M6	online / continuous	RQ5/RQ6 only after offline finite-candidate evidence

Table 3: Milestones from current docs; current focus is M1/M2 scale readiness before target rollout scale-up.

Immediate next edits before scale

- Fix all-hard-diagnostic invalidity consistency.
- Add production preflight JSON and stale-schema checks.
- Require scene-level split manifest before shard grouping.
- Retune three-family sampler after per-family validity reports.
- Reduce Zarr chunk/file bloat.
- Validate $H=1$ target-label profile, then $H>1$ rollout traces.

What I Need Feedback On: RQ Locks

Decisions to lock

- 1 Does RQ1 lock objective, fixed-budget evaluation, root-normalized training return, and offline fitted Q_H as first method?
- 2 What pass/fail bar defines positive offline Q_H recovery?
- 3 Which RQ3 ablations are mandatory before Q_H : semidense+DINO, target crop, $\mathbb{S}^2$ memory, candidate-query pooling?
- 4 Does RQ4 cover invalidity, candidate diversity, support scale, and the external-data caveat clearly enough?

Evidence protocol to lock

- Scene-level split and acceptable scale fallback.
- Target matching thresholds and ambiguity policy.
- Invalidity as hard masks/reasons.
- LRZ/Zarr preflight before broad generation.
- RQ5/RQ6 extension wording and optional Q_H -as-critic bridge.

Meeting goal: align scope, RQs, and next gates before more rollout data is generated.